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Systems, methods, and computer-accessible mediums for repressing or turning off the read of a digital tag

Abstract

An exemplary system, method, and computer-accessible medium can include, for example, storing on a first device a digital tag, the tag configured to be associated with at least one application on a second device, such that receipt of the tag on the second device launches the application on the second device; and emitting from the first device to a second device the digital tag; wherein the application on the second device is configured to launch in a special state when the second device is in at least one state from a pre-determined list of states.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The subject application is a continuation of U.S. patent application Ser. No. 17/477,307, filed Sep. 16, 2021, now U.S. Pat. No. 11,743,370, which is a continuation of U.S. patent application Ser. No. 16/806,702, filed Mar. 2, 2020, now U.S. Pat. No. 11,153,428, the complete disclosures of which are incorporated herein by reference in their entireties.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to repressing or turning off the read of a digital tag, such as an NFC tag, by a user device, such as a smartphone (e.g. an iPhone® or Android® phone).

BACKGROUND INFORMATION

(2) Near-field communication (NFC) enabled devices, such as smartcards and other devices, have become ubiquitous and continue to increase in popularity. Similarly, smartphones have also become more and more popular in recent years. Data contained on an NFC-enabled smartcard can be read when an NFC field is brought sufficiently close to the NFC smartcard. This energizes the NFC components on the smartcard and allows for information to be read from the smartcard. Many smartcards are configured to allow access to the encoded information by any suitable receiving device. This can lead to security problems, particularly in the context of a mobile device capable of reading data from a contactless chip. Mobile devices configured with, for example, an NFC reader are capable of reading credit card information from a card chip. This can allow the mobile device to read the information on the card whenever the credit card is within the range of the NFC field of the mobile device.

(3) Certain operating systems are configured to cause a smartphone or other device to continuously operate its NFC field, allowing the smartphone to read any NFC smartcards that are brought close to the smartphone. As an example, devices operating the Android® operating system by Google® are configured to have an NFC field which is constantly active. This can result in unintentional reads of NFC-enabled smartcards when the smartcard is brought close to it. For example, when NFC-enabled cards are placed in a user's pocket together, or when an adhesive wallet which is attached to the back of a phone to store cards is used, smartcards can be read by a device which the user does not intend for the smartphone, or other device, to read the information contained on the smartphone or other device.

(4) In other cases, the smartphone may, responsive to receipt of the NFC tag, launch an application or perform another action. This can be contrary to a smooth user experience in use of the smartphone as the user may not wish that an inadvertent tap of a smartcard launch an application. In other cases, due to being in close proximity when carried by a user (such as in a user's wallet), multiple NFC tags might be read, causing more than one application to be launched, or actions taken, without the user intending for these actions to be performed. Simultaneously, a user may not wish to disable the NFC capabilities of his or her device due to the convenience of using an NFC-enabled smartphone and NFC-enabled devices.

(5) In yet other cases, a counter on a smartcard can increment each time the NFC-enabled smartcard is read. However, the counter may not necessarily correspond to an independent counter maintained at the issuer of the smartcard. The reason for the difference in correspondence can occur because, for example, the counter on the smartcard may be incremented each time the smartcard is read, but the read or information being read from the smartcard may not be sent to the issuer of the smartcard. This difference can lead to security issues when the two counters are not synchronized. Accordingly, there is a need to maintain synchronization of the counter on the NFC-enabled smartcard without disabling the NFC capabilities of the smartcard.

(6) Conventional methods related to NFC and the near-field data exchange format (NDEF) have limitations that do not allow for the functionality of an NFC tag to launch an application, prevent unintentional launches, and/or allow for a user to toggle the functionality of, for example, an NFC smartcard. These are other deficiencies exist. Thus, it can be beneficial to provide exemplary systems, methods, and computer-accessible mediums that selectively repress or turn off the read of an NFC tag by a smartphone. It can be further beneficial to provide exemplary systems, methods, and computer-accessible mediums to toggle the functionality of repressing or turning off the read, or launching of an application responsive to the read of an NFC tag, by a smartphone. Even further, it can be beneficial to provide exemplary systems, methods, and computer-accessible mediums, which can synchronize (and/or maintain the synchronization of) the counters on the smartcard and the issuer of the smartcard.

SUMMARY OF EXEMPLARY EMBODIMENTS

(7) Embodiments of the present disclosure provide a system, method, and computer-accessible medium, which can include, for example storing on a first device a digital tag, the tag configured to be associated with at least one application on a second device, such that receipt of the tag on the second device launches the application on the second device; and emitting from the first device to a second device the digital tag; wherein the application on the second device is configured to launch in a special state when the second device is in at least one state from a pre-determined list of states.

(8) Embodiments of the present disclosure provide a system, method, and computer-accessible medium, which can further include, for example: receiving from a first device digital information upon physical proximity with the first device; receiving from the first device a digital tag configured to be associated with an application; launching, responsive to the tag, the application associated with the received digital tag; and for a fixed time duration after the receipt of the digital tag, through the application, intercepting additional digital information from the first device when the computer hardware arrangement is in at least one state from a pre-determined list of states.

(9) Embodiments of the present disclosure provide a system, method, and computer-accessible medium, which can further include, for example: emitting from the first device a digital tag, wherein the tag is configured to be in the near-field communication data exchange format specification, further configured to be an Android Application Record launch tag and further configured to be associated with at least one application on a second device, such that receipt of the tag on the second device launches the application, wherein the application is configured to intercept additional information emitted from the first device within a fixed time period when the second device is in at least one state from a pre-determined list of states.

(10) Further features of the disclosed design, and the advantages offered thereby, are explained in greater detail hereinafter with reference to specific example embodiments illustrated in the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating a first device according to an example embodiment of the present disclosure.

(2) FIG. 2 is a diagram illustrating a user device according to an example embodiment of the present disclosure.

(3) FIG. 3 is a flow diagram of a method for launching an application on a device according to an example embodiment of the present disclosure.

(4) FIG. 4 is a flow diagram of a method for launching an application on a device and synchronizing counters according to an example embodiment of the present disclosure.

(5) FIG. 5 is a flow diagram of a method for launching an application on a device and synchronizing counters according to an example embodiment of the present disclosure.

(6) FIG. 6 is a flow diagram of a method for modifying digital tags of a smartcard according to an example embodiment of the present disclosure.

(7) FIG. 7 is an illustration of a block diagram of an example system in accordance with example embodiments of the present disclosure.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

(8) The following description of embodiments provides non-limiting representative examples referencing numerals to particularly describe features and teachings of different aspects of the invention. The embodiments described should be recognized as capable of implementation separately, or in combination, with other embodiments from the description of the embodiments. A person of ordinary skill in the art reviewing the description of embodiments should be able to learn and understand the different described aspects of the invention. The description of embodiments should facilitate understanding of the invention to such an extent that other implementations, not specifically covered but within the knowledge of a person of skill in the art having read the description of embodiments, would be understood to be consistent with an application of the invention.

(9) An example system, method, and computer-accessible medium can be provided which can facilitate the blocking or repressing of a device of the read of information contained on a smartcard by for example a computer or smartphone. The smartphone can be an Android® phone, but any smartphone having the functionality described in this disclosure is within the scope of the invention.

(10) A smartcard can be for example, an identity card, which can be read by a mobile device. For example, information encoded on a user's smartcard can be read by a mobile device, or other similar device, upon physical proximity of the smartcard to the device. A smartcard is often encoded to provide digital information to the device without any configuration—accordingly, the

information on the card can be read by any piece of software installed on the digital device.

However, a smartcard can be encoded to include, for example, a digital tag. When the smartcard is in proximity to a device, the digital tag can first be read by the mobile device. Generally, upon receipt of the digital tag, an application identified by the digital tag, or otherwise configured to be associated with the payload in the digital tag, is launched automatically.

(11) The digital tag can be configured to, when read by a mobile device, first launch an application identified by the digital tag. An application is typically launched responsive to receipt of a digital tag. In addition, and if necessary, the digital tag can be configured to prompt the mobile device to download and install an application prior to launching it. In some examples, the digital tag can be configured to cause the mobile device to launch the application store associated with operating system running on the mobile device (e.g., the Google® Play Store for Android® devices) to facilitate the download and installation of the application.

(12) In some examples, the mobile device can continuously monitor for the receipt of digital tag(s) on the mobile device. The mobile device can further contain a background service that can continuously monitor or otherwise capture information about the state of the mobile device. In some examples, the background service can be part of the application identified by the digital tag. Upon receipt of the digital tag by the mobile device and in combination with information about the state of the mobile device collected by the background service (e.g. whether the screen is on, whether the device is locked/unlocked, whether the power saver is on/off, what applications are running, what tasks are pending completion, whether a communication interface is engaged/disengaged, whether or not the device is awaiting a user input, whether the processor is operating above or below a certain level, whether the battery is above or below a certain level, or if the target application has already been launched or is engaged in a task), the mobile device can launch the application associated with the digital tag into a non-visible state. For example, if the digital tag is read within a threshold time after the mobile device screen unlocks (e.g., within 100 milliseconds, within 200 milliseconds, or within 400 milliseconds), the monitoring application can determine this to be an inadvertent read and thus, the application would not be launched. Inadvertent reads of digital tags can be caused for a number of reasons, including without limitation, including proximity to an emitting device and the activation of the mobile device's radio receiver when its screen is unlocked by a user.

(13) Further, if the mobile device detects another tap or receipt of the same digital tag within a predetermined period of time, the mobile device can then launch the application associated with the digital tag in a visible state for the user to interact with. In addition, if the application is launched into a non-visible state responsive to receipt of the digital tag, the mobile device can also read additional information from the manifest of multiple tags stored on a smartcard, and send the payload of such tags to a web service. If an additional read of a digital tag is not present within a pre-determined period of time, the application can exit, without the user of the device ever seeing an interface.

(14) An application can also be configured to, either automatically or based on user input or preferences, write to an applet located on a smartcard which is compliant with JavaCard®, or another smartcard operating system. An applet on the smartcard will generally generate an NFC compatible tag, such as an NDEF tag, whenever the smartcard is powered by the field of the smartphone. The application can thus toggle the functionality of the smartcard to turn off the creation of NFC tags, or otherwise modify the smartcard, so that a user can choose whether he wants the smartcard's NFC functionality to be enabled or disabled.

(15) The launched application can for example be configured to prevent other applications on the smartphone device from accessing the requested information. The application associated with the digital tag can be installed on the device by the user. The tag can be generated in the NDEF data format specification. The NDEF tag can further be configured as an Android® Application Record (AAR) launch tag.

(16) The application associated with the digital tag can further be configured to perform a variety of functions. For instance, the application can send a notification to a user when the application has been launched more than a pre-determined number of times in a pre-determined period of time. The application can also create a list of other applications installed on the device. Any other functionality generally associated with an application on an operating system, such as for example, an application on the Android® operating system, can be programmed into the device.

(17) FIG. 1 is a diagram illustrating a first device **100**. The first device can be any device capable of storing information such that the stored information can be accessed. First device **100** can be capable of storing a record **101**. The first device can also have a processor **102**. Processor **102** can be any suitable commercially available or custom designed processing circuitry. The first device can also have a communication interface **103** capable of generating a first device communication field **104**. The first device can also have memory **105**. The first device **100** can be any device capable of containing a record **101** and able to communicate through a communication interface **103**, including a payment card, credit card, a debit card, a user identity card, a cell phone, a smart phone, an RFID card, a tablet, a computer, an NFC card, or any other communication device. It is understood that the present disclosure is not limited to a particular type of card or first device, and includes any type of card or first device capable of performing the functions described herein.

(18) The first device **100** can also be a smartcard, with any suitable smartcard operating system stored on the first device **100**. For example, the operating system can be stored on memory **105**. The card operating system can be run using the processor **102** of first device **100**. Two examples of operating systems available for smart cards include JavaCard® and the MULTOS Card Operating System®. Card operating systems can allow for the on-card application development of programs that need to run within the secure environment of the smart card chip—such as for example, chip enablement, issuance, controlling files, and managing data loads. Specific applications can be installed on the card operating system to for example, generate NDEF tags that follow certain specified parameters. These generated tags can be stored in record **101**.

(19) Record **101** can include any information that is accessible by a suitable computing device. Record **101** can be stored within memory **105**. Record **101** can be stored in any suitable data type, including Boolean, byte, cbyte, date, decimal, integer, long, number, string, or any combination of those data types. Record **101** can also be stored as delimited, fixed, or mixed data. Record **101** can contain multiple pieces of data, which all can have corresponding meta-data. The records can also include custom data types or data fields. Non-limiting examples of data can include information such as the first device's identifier number, a credit card number, a personal account number (PAN), a user name associated with the first device, an expiration date for the first device, and a card verification value (CVV) code.

(20) Record **101** can also be stored in a manner to allow for data to be exchanged over the NFC Data Exchange Format (NDEF), which allows for the exchange of information between any compatible NFC device and another NFC device or tag. NDEF is strictly a message format. It is a binary message encapsulation format to allow for the exchange between NFC-enabled devices. NDEF messages can contain payloads of arbitrary type and size. The NDEF data format can store be used to store and exchange information such as Uniform Resource Identifier (URIs), plain text, etc., using a format that is commonly understood. The NDEF data format further can further support the exchange of NDEF messages as a mechanism to allow for the exchange of NDEF records. Each NDEF record can contain a structure that identifies the content of the record as well as the size of the record. An NDEF record can contain on a basic level two components, a record type used to provide context to a payload data and a payload data. Together, these two components represent the action to be taken by a device upon receipt of an NDEF record. One NDEF message can contain more than one NDEF record. The NDEF file itself can be created dynamically. For example, the NDEF file can be based on a random number generator or externally introduced data. An NDEF file is read when a card, or a chip contained in the card is exposed to a properly aligned

magnetic field and a request is issued for a specific NDEF message.

(21) Record **101** can further be stored as an AAR launch tag. An AAR launch tag is a special type of NDEF record, which is used by Google's Android® operating system, to signify to an NFC device that an explicitly defined Android® Application should be used to handle the NFC tag. An AAR launch tag can include a package attribute, which can identify which Android® application will handle or process the NFC tag. The package attribute can identify which Android® application will be launched in response to the tag.

(22) Record **101** can also be stored in a format compatible with the iOS operating system framework. For example, the iOS operating system allows for the use of the Core NFC library to allow for Near Field Communication (NFC) tags of types **1** through **5** to be read, which contain data in the NFC Data Exchange Format.

(23) Memory **105** can be a read-only memory, write-once read-multiple memory or read/write memory, e.g., RAM, ROM and EEPROM, and a first device **100** can include one or more of these memories. A read-only memory can be factory programmable as read-only or one-time programmable. One-time programmability provides the opportunity to write once then read many times. A write once/read-multiple memory can be programmed at a point in time after the memory chip has left the factory. Once the memory is programmed, it cannot be rewritten, but it can be read many times. A read/write memory can be programmed and re-programmed many times after leaving the factory. The read/write memory can also be read many times. The memory of user device **200** can be programmed with instructions to generate a record **101**. Record **101** can further include digital tags, which can be generated based on instructions included in memory **105**. The memory of user device **200** can also include instructions stored as specialized applets, such as a JavaCard® applet. For example, memory **105** can generate one or more NDEF tags, which can further be configured to be AAR NDEF records associated with a particular Android® application. Memory **105** can thus advantageously be modified if needed to change which Android® application on an Android® device responsive to receipt of a tag. In another example, memory **105** can be configured to generate a digital tag which is configured to be compatible with a different operating system or different application.

(24) Memory **105** can store other information, such a user identifier. Record **101** can also be included in memory **105**. A user identifier, algorithms, other information such as cryptographic keys can also be stored in memory **105**. Memory **105** can be divided into several zones, with each zone having a different level of security. The processor **102** can keep track of which memory addresses belong to which zones and the circumstances under which each zone can be accessed. In an example embodiment, the memory **102** can be divided into four zones: a secret zone, a confidential zone, a usage zone, and a public zone.

(25) A secret zone can be used for storage of information which can be used only by processor **102**, e.g., passwords, cryptographic keys. The information stored in this zone is not readable outside of the first device **101**. The zone can also contain a dynamic algorithm which determines the tags that can be sent by a transfer device. In an example, the secret zone can be implemented with a separate processor that is capable of performing cryptographic functions. Cryptographic keys can be passed in to the secret zone or can be generated in the secret zone, and in either case the keys can be stored in the secret zone and used to support cryptographic services. If necessary, cryptographic keys can be exported from the secret zone. Cryptographic keys and passwords can also be stored directly in memory **105** without the use of a secret zone.

(26) Communication interface **103** can be of any suitable technology capable of sending or receiving data over a distance. Examples of such technology include, for example, Wi-Fi, WLAN, RF, radio, IR, Bluetooth, RFID, NFC or any combination thereof or any other appropriate architecture or system that facilitates the communication of signals, data, and/or messages. Similarly, any suitable hardware level and software level algorithm can be chosen to allow for the transfer of data on a first device communication field **104**. The first device communication field

104 can be generated by communication interface **103** using one or more communication protocols, including the asynchronous connection-less protocol, synchronous connection-oriented link, link management protocol, host controller interface, or low energy link layer. Information can be communicated using the NDEF data exchange format. The use of this format allows for the exchange of both NDEF messages and NDEF records. The NDEF data format can support NDEF messages that are a mechanism allowing for the exchange of NDEF records. Each NDEF record can contain a structure that identifies the content of the record as well as the size of the record. An NDEF record can contain two components, a payload data and a record type used to provide context to the payload data. Together, these two components represent the action to be taken by a device upon receipt of an NDEF record.

(27) FIG. 2 is a diagram illustrating a user device **200**. Non-limiting examples of user device **200** include any NFC reading device, such as a personal computer, a laptop, a tablet, a cell phone, an NFC reader, or a smart phone. It is understood that the present disclosure is not limited to a particular type of user device, and includes any type of user device capable of performing the functions described herein. User device **200** can include a display **201**, a user interface **202**, memory **203**, a processor **204**, and a user device communication interface **205** with a user device communication field **206**. User device **200** can further be configured to contain one or more operating systems, such as for example, the Android® operating system.

(28) Memory **203** can be a read-only memory, write-once read-multiple memory or read/write memory, e.g., RAM, ROM and EEPROM, and a user device **200** can include one or more of these memories. A read-only memory can be factory programmable as read-only or one-time programmable. One-time programmability provides the opportunity to write once then read many times. A write once/read-multiple memory can be programmed at a point in time after the memory chip has left the factory. Once the memory is programmed, it cannot be rewritten, but it can be read many times. A read/write memory can be programmed and re-programmed many times after leaving the factory. It can also be read many times.

(29) Memory **203** can further be configured to be installed with an operating system or special instructions. The installed operating system can further be configured to install additional compatible pieces of software, such as applications or programs. Examples of commercial operating systems include for example the Android® operating system, iOS®, Windows®, Debian®, Linux®, and Linux® variations such as Ubuntu®. Special purpose operating systems can also be installed on memory **203**. As another example, more than one operating system can be installed on memory **203**, allowing a user of the user device **200** to select which operating system the user wishes to use.

(30) Display **201** can be selected from any suitable two-dimensional or three-dimensional display, such as a light-emitting diode, liquid crystal display, digital light processing display, or organic light-emitting diode display. User interface **202** can be selected from any suitable user input device such as a touchpad, a touchscreen, a mechanical switch, natural language user interface, a click-wheel, QWERTY keyboard, mouse, gesture recognition, or capacitive touchscreen.

(31) User device communication interface **205** can be of any suitable technology capable of sending or receiving data over a distance. Examples of such technology include, for example, Wi-Fi, WLAN, RF, radio, IR, Bluetooth, RFID, Near-Field-Communication (NFC) or any combination thereof or any other appropriate architecture or system that facilitates the communication of signals, data, and/or messages. Similarly, any suitable hardware level and software level algorithm can be chosen to allow for the transfer of data on user device communication field **206**. User device communication field **206** can be generated by user device communication interface **205**. Examples of algorithms include the asynchronous connection-less protocol, synchronous connection-oriented link, link management protocol, host controller interface, or low energy link layer. Information can be communicated using the NDEF data exchange format. The use of this format allows for the exchange of both NDEF messages and NDEF records. The NDEF data format can support NDEF

messages that are a mechanism allowing for the exchange of NDEF records.

(32) First device **100** and user device **200** can also communicate with one another using a network. Network can be any means, mechanism, protocol, or pathway to allow information to be transferred between the first device **100** and user device **200** including but not limited to Wi-Fi, WLAN, RF, radio, IR, Bluetooth, cellular, near-field communication, SMS, MMS, a phone call, any combination thereof, or any other suitable architecture or system that facilitates the communication of signals, data, and/or messages. The first device and the user device can be in communication through a single network or a multitude of networks.

(33) In an example embodiment, first device **100** can become energized when it comes close to an NFC reader which can be contained on user device **200**. The NFC reader can emit a magnetic field which can energize first device **100** (e.g. a smartcard) through inductance. The first device **100** (e.g. a smartcard) can also generate a manifest of tags through stored programming, such as an applet, which dynamically generates NFC tags, such as a random number. The first device, once powered, through this or another method, can then emit an NFC signal. Upon detecting the NFC signal emitted by first device **100**, user device **200** can automatically attempt to read an NDEF message by sending an NDEF request to first device **100**. First device **100** can accept the request from user device **200** and respond with an NDEF message. The NDEF message can further contain at least one record that is an AAR launch tag. Upon first device **100** and user device **200** being within physical proximity of each other, the user device can request information from the first device. The first device can in response to the request, transfer the request to user device **200**. The first device can also be configured to emit information automatically to any user device **200** capable of receiving information from first device **100**. This can be done, for example, when first device communication field **104** overlaps with user device communication field **206**. This transfer of information between the first device and the user device can be accomplished through the means described above.

(34) The first device can be configured to include for example, information such as a credit card number, and associated information, such as the CVV number, user name, and expiration date. This first device **100** can further be configured to transfer this information to user device **200** through the use of NFC, or a similar medium. This information can be accessed by any second device upon physical proximity of the two devices. As explained above, upon receipt of the NFC tag, an application installed or otherwise associated with the NFC tag, can be launched by user device **200**. The application will typically be launched in a visible mode on user device **200**, with full interactivity. The inadvertent launch of one or more applications can interfere with a user's intended use of the NFC functionality of a first device **100** and user device **200**.

(35) However, user device **200** can further be configured to include a background service. This background service can be installed in tandem with the installation of the application associated with a tag emitted from first device **100**. In some examples, the background service can be part of the application associated with the tag emitted from the first device **100**. The background service installed on user device **200** can monitor for the state in which user device **200** is in. For example, the background service can monitor for the state of the device, such as whether the device screen is active, the screen is locked or unlocked, the battery or power of the device, and timing between certain actions (e.g., a time threshold between screen unlock and tag read). The background service can also monitor for whether the application associated with the digital tag has already been launched. Upon receipt of a digital tag, the mobile device, upon detecting the tag associated with an application, and upon review of the appropriate state of the device as collected by the background service, can cause an application associated with the received digital tag to be launched. The background service can then cause the launch of an application associated with the digital tag received by user device **200** in a special state, such as a non-visible state, where the user cannot see or interact with the interface of the application. Thus, the user is not hindered by the launch of an application when he or she does not intend for that application to be launched.

(36) As an example, the background service can act to stop the launch of the application when the user device **200** is within communicating proximity with the first device **100** and a digital tag is read by the user device **200**. The state where the user device **200** screen is locked and the application is launched upon a reading of the digital tag can be considered undesirable, in contrast to the state where a user has intentionally caused the user device **200** to read the digital tag. The background service can monitor the time difference between the screen of user device **200** unlocking and the read of the digital tag by user device **200** and compare this difference to a threshold. If the difference is less than the threshold (i.e., if the time between the screen unlock and the tag read is too small) then the launch of the application can be stopped. If the difference is greater than the threshold (i.e., if the time between the screen unlock and the tag read is sufficiently large) then the launch of the application can proceed.

(37) As a further example, the background service can monitor and track the time every time a certain event occurs, such as for example, a user unlocking the screen of user device **200**. Thereafter, the background service can record the time at which a digital tag, such as an NFC tag, is received on user device **200**. The background service can then compare the two times, and if the times are within a threshold, such as for example, a short period of 200 milliseconds, the background service can be configured not to launch the application associated with the NFC tag in a visible state, and rather launch the application in a non-visible state. In this state, other information received by user device **200**, such as for example, the remainder of the payload encoded in an NDEF format, can be stored in memory or storage by user device **200**. This information can then be read by the application (which is still in a non-visible state) and acted upon. For example, this information can be sent to a web service. This feature can be advantageously used to synchronize the read count between a smartcard (e.g. first device **100**) and an entity issuing the smartcard (e.g. an agency, bank, or other entity).

(38) In another example, a time tracking can be utilized as a guard against execution of the payload. For example, the background service can start to record the time after an event, e.g., powering on the user device **200** or a user unlocking the screen of the user device **200**, and the background service can prevent the execution of NDEF payload for a threshold period of time, e.g., 10 seconds, from the time of the event. As another example, the time tracking can commence upon receipt of a digital tag by the user device **200**.

(39) In another embodiment, first device **100** can further be configured to include a tag in its record **101**, such as an AAR launch tag, appended to the information typically sent by the first device. The AAR launch tag can be configured to, once received by the user device, launch an application specified by the AAR launch tag. This tag can be configured to first be sent to user device **200**, before any other tag contained in first device **100** is transferred. This application can be configured to only allow an application associated with a second tag to be launched when the first tag and the second tag are received by user device **200** within or outside a pre-determined period of time. For example, a typical period of a 100 milliseconds can be used, which may correspond to typical human reaction time in tapping a card to a user device. If a same pair of tags are received within this period, it is likely that the card is within proximity of the device (such as on a wallet attached to the phone) and that the user did not intend for the user device to respond to the tag. However, if the period of receiving the tags is within a certain time frame, the background service or application can allow the application to be launched in its full state, which is visible to the user of user device **200**.

(40) First device **100** can also be configured to include or generate a collection of tags, to allow for this process of launching an application to be compatible across operating systems and user devices.

(41) As another example embodiment, an NFC tag contained on a smartcard (e.g. first device **100**) can be selectively toggled or turned on or off by a user that has a user device (e.g. user device **200**). As an example, user device **200** can contain an application which is able to write to first device **100**

through any of the communication methods described above. For example, user device **200** can communicate with first device **100** through NFC. User device **200** can contain an application which allows the user of user device **200** to toggle (that is, turn on or off, or modify the behavior of) of the signals sent from first device **100**. This can be accomplished, for example, by an application on the user device which displays the NFC options that are contained on a particular smartcard. For example, the application can display the name of the smartcard, some identifying information about the smartcard (such as for example the issuer of the smartcard, the name on a smartcard, or a number on the smartcard) along with the digital tags that a user may wish to turn on or off. The application can, through the use of NFC or other communication medium on user device **200**, then communicate with, and in turn write to the memory of first device **100**. The application can then, for example, write to an applet that is located on, for example, a JavaCard® compliant card. By writing to or otherwise modifying this applet, the applet contained on first device **100** can be modified to not generate an NFC compatible tag, such as an NDEF message, whenever the first device **100** is energized by user device **200**, through for example, a magnetic field. The application on user device **200** can accomplish this by writing to the applet on the smartcard disabling the functionality that generates the tag. In some examples, the determination to turn on or turn off reading of the tag can be done for security purposes and based on, e.g., the state of the device, location data, time, and/or other criteria.

(42) Additional information to verify that only an authorized change is being made to the first device **100** can include security algorithms on either or both user device **200**, the application, and first device **100**. For example, a secret code or hash can be sent between first device **100** and user device **200** when such a change is attempted.

(43) As an example, a user may keep their smartcard (such as first device **100**) to the back of their Android® phone (such as user device **200**) or store their card in a phone wallet immediately adjacent to the phone's NFC reader. An application may be installed on the phone that has been programmed to read an Europay®, MasterCard®, and Visa® (“EMV”) applet on the card could read the credit card number, personal information, CVV code and expiration date from the card. In addition to this, the NFC reader, upon receiving a digital tag, such as for example, an NDEF record, or an AAR launch tag, can launch an application.

(44) The addition of the NDEF records to the smartcard does not prevent applications that have already been launched by the user on user device **200** from accessing or requesting information on the first device **100**. Applications on the Android® operating system can be automatically launched in the presence of events, such as the presence of an NFC signal. When no application is active on user device **200** however, the NDEF record ensures that the first party application is launched on user device **200** responsive to receipt of the NDEF tag. The Android® operating system is configured to only allow one application to access information from an NFC signal at a time. This can occur when, for example, user device **200** is in a locked state, or when the screen of user device **200** is inactive. However, when an application is already launched by the user of user device **200** is not inhibited from accessing or using that application in the manner intended by the user.

(45) As an example, a user can have his or her NFC-enabled card in his or her purse or wallet. Throughout the course of the day, the enabled card can come into proximity with the user's personal user device, or with other NFC-enabled user devices, such that the card can be energized and read. For example, the user can have a device with the Android® operating system installed on his or her device. The first device can further be configured to include a tag in its record **101**, such as an AAR launch tag, configured to launch for example, a Capital One® application on the Android® phone. If the application is not already installed on the device, the tag can further be configured for the Android® phone to launch the Google® Play Store, or another similar platform, to allow for the designated application to be installed.

(46) The NDEF tag or AAR launch tag is configured with the package name of a specific Android® application. Thus, upon receipt of the particular AAR launch tag or NDEF tag, the operating

system, or the Android® operating system, will launch the specified application when the contactless card is tapped to the phone. If the specified application is not currently installed on the Android® device, it may cause the Google® Play Store to be launched to the page where the specified application can be downloaded and installed onto the Android® device.

(47) The tag in the first device **100**, can further be configured to cause the second device **200** to access a server to download the application associated with the tag in first device **100**. This can be accomplished by including information identifying the particular application in the first device **100**. The server can be a standard server well-known and commercially in use to communicate over a network with a device. In one example, the tag in the first device can cause a specialized application to interface with the server and cause the application to be downloaded. One example of such a specialized application is the Google® Play Store on Android® operating system. Other examples of such applications include the App Store on the iOS operating system and the Microsoft Store allowing Windows® applications to be downloaded. Yet another example is the Canonical Store and the Ubuntu Software Center on the Ubuntu® operating system. Other specialized package managers exist for various operating systems.

(48) In an example embodiment, a backend service can be informed of each read of the EMV chip contained in first device **100** or of each NFC read. This can occur through the application associated with a digital tag emitted from first device **100** reading the payload when launched in a non-visible state. In this manner, the provider of the first device **100**, such as the provider of a smartcard or credit card, can be made aware each particular use of the first device **100**. For example, a credit card provider can be informed of each read of the EMV chip. Thus, the credit card provider could match information of each read of the EMV chip with that of known payments made by the customer. In this manner, the credit card provider could use the comparison for fraud protection. In another example, the issuer of a smartcard can be made aware of each read from the EMV chip, and compare each known read of the EMV chip with all attempts to access the information contained in the EMV chip. This information can be used to collectively synchronize a counter on a smartcard such as the first device and a backend service.

(49) FIG. 3 is a flow chart diagramming a method **300** of transferring information from first device **100** to user device **200** in an example embodiment of the present invention. The method commences at step **305**, in which an applet or digital tag is stored on a first device **100**. This applet can be configured to generate a collection or manifest of digital tags. These tags can be pre-generated, in which case, an applet can optionally not be used to generate the digital tags. In step **310**, a signal is emitted from the first device. This can be emitted in response to an external trigger, such as an electronic field, or first device **100** can be in a state of constant emission of the signal. First device **100** can receive a request for information. This step can for example take the form of the NFC apparatus in first device **100** being activated in the presence of an external field or device. This can also occur when first device communication field **104** overlaps with another communication field. For example, an NDEF file is read when the card/chip is exposed to a properly aligned magnetic field and a request is issued for a specific NDEF message. In step **315**, first device **100** can emit a digital tag. In this step, first device **100** can transmit additional information along with the digital tag. Examples of additional information sent along with the digital tag can include user information or credit card information. In step **320**, the emitted tag can be configured in such a manner that the emitted tag can be associated with at least one of an application such that receipt of the emitted tag on the second device launches the associated application on the second device. The tag can also be further configured to be associated with an input on the second device. An input can also be a particular state of a user device (e.g. user device **200**) in which the device is in a particular state, such as the user device being unlocked, the screen of the user device being active, or the screen being touched. The configuration of the tag can also be done in a manner which corresponds with the NDEF format or the AAR format. The AAR format is compatible with Android® operating system devices and can specify a particular action

for the Android® operating system to take upon receipt of the AAR launch tag. For example, it can specify a particular application to launch in response to the received tag.

(50) FIG. 4 describes one embodiment of the present invention, method **400**. The method commences at step **405**, in which an applet or digital tag is stored on a first device **100**. This applet can be configured to generate a collection or manifest of digital tags. These tags can be pre-generated, in which case, an applet can optionally not be used to generate the digital tags. In step **410**, a signal is emitted from the first device. This can be emitted in response to an external trigger, such as an electromagnetic field, or first device **100** can be in a state of constant emission of the signal. First device **100** can receive a request for information. This step can for example take the form of the NFC apparatus in first device **100** being activated in the presence of an external field or device. This can also occur when first device communication field **104** overlaps with another communication field. For example, an NDEF file is read when the card/chip is exposed to a properly aligned magnetic field and a request is issued for a specific NDEF message. In step **415**, first device **100** can emit a digital tag. In this step, first device **100** can transmit additional information along with the digital tag. Examples of additional information sent along with the digital tag can include user information or credit card information. In step **420**, the emitted tag can be configured in such a manner that the emitted tag can be associated with at least one of an application such that receipt of the emitted tag on the second device launches the associated application on the second device. The tag can also be further configured to be associated with an input on the second device. An input can also be a particular state of a user device (e.g. user device **200**) in which the device is in a particular state, such as the user device being unlocked, the screen of the user device being active, or the screen being touched. In step **425**, the emitted tag can be configured to launch an application on the second device. This application can be launched for example when the emitted tag is received, and the payload processed by a user device (e.g. user device **200**) which in turn causes the application to be launched on the user device. In this step, a background service can record, for example if the screen is on or off, and based on that, and/or receipt of the emitted tag, start a counter. For example, a background service can record the time when the screen of the user device is turned on. However, the background service need not be started at this step and can be started independently of this step. Step **430** consists of configuring the application associated with the digital tag to be launched into a non-visible state. In this state, the application can be running but would not be visible to a user. Step **435** can consist of configuring the application such that launch of the application can be in a full visible state on a user device (e.g. user device **200**). This step can further consist of monitoring the time or stopping the time recording started by the background service when the user device receives the emitted tag by the background service (e.g. an Android® Broadcast Receiver). The time can be used as described to determine whether the application associated with the digital tag should be launched in a visible state.

(51) FIG. 5 describes an embodiment of the present invention, method **500**. Method **500** starts at step **505**, in which information is requested from a first device (e.g., first device **100**) when a user device (e.g., user device **200**) is in physical proximity to a first device. This can occur when user device **200** recognizes the presence of a first device through the overlap of communication fields of first device **100** and user device **200** overlap. This can also occur through software in user device **200** recognizing the presence of a communication field external to user device **200**. In step **510**, a digital tag is received from first device **100** (e.g. first device **100**). This digital tag can be information which was not requested from first device **100**. This digital tag can be platform or operating system specific. In some embodiments, this digital tag can be configured to be an AAR launch tag. In step **515**, an application associated with the emitted digital tag can be launched responsive to the received tag. For example, on an Android® operating system, the launched application can be defined by meta-data or specialized fields in the received tag. The second device can, upon not finding the application specified in the emitted digital tag, launch a package manager

or application manager, such as for example, the Google® Play Store, in order to identify and allow the user to download the specified application. In step **520**, any additional tags received by the user device can further be intercepted. In step **525**, any additional digital information received from the first device can be intercepted when a computer hardware arrangement (e.g. user device **200**) is in a particular state. For example, if a cell phone has its screen on, it can then be considered to be in state amongst the list of pre-determined states. This can include for example one or more tags associated with, or sent subsequent to the first digital tag, that instructs the device receiving the tags to perform certain actions (e.g. launch an application) or otherwise delivers a payload to the second application. In step **530**, the application launched in response to receipt of a digital tag from the first device can send instructions to the first device. These instructions can disable, overwrite, or de-activate one or more digital tags on the first device that are associated with at least one application.

(52) FIG. **6** describes an embodiment of the present invention, method **600**. Method **600** can begin at step **605**, in which at least one digital tag is stored on a first device (e.g., first device **100**). An applet can also be stored on the first device, which can further be configured to generate a collection or a manifest of digital tags. These tags can be pre-generated, in which case, an applet can optionally not be used to generate the digital tags. The applet can also be modifiable or configurable, such as through the use of flags, which can indicate which digital tags the applet can generate. In step **610**, a digital tag is emitted from the first device (e.g., first device **100**). This can be emitted in response to an external trigger, such as an electromagnetic field, or first device **100** can be in a state of constant emission of the signal. This step can for example take the form of the NFC apparatus a first device (e.g. first device **100**) being activated in the presence of an external field, such as a magnetic or electric field, from a second device (e.g. user device **200**), inducing power within first device **100**. The NFC apparatus can also be activated when the first device communication field (e.g., first device communication field **104**) overlaps with another communication field, such as that belonging to a second device. In step **615**, the tag emitted from the first device (e.g. first device **100**) can be configured to be associated with one at least one application, and further be configured to cause the application to be launched on a second device (e.g. user device **200**). This can be done by configuring the emitted tag to be in a particular data format, with a specific payload. For example, the digital tag can be configured to be in the NDEF data format specification, further configured to be an AAR launch tag. In step **620**, the application associated with the emitted tag can be configured to only launch in either a visible or non-visible state depending on information received from a background process. This background process can for example be started whenever a second device (e.g. user device **200**) is in a particular state, such as for example, the screen of the second device is on, the second device has received a user input, or the second device is unlocked, or the second device is turned on. In step **620**, the application associated with the digital tag can launch, or caused to be launched, in a visible state. In this state, a user can interact with the application on a second device (e.g. user device **200**). For example, a user can interact with the second device using the touchscreen of the second device. In step **625**, the application can further be configured to include an option to configure the digital tags included on a smartcard (e.g. first device **100**). As an example, a smartcard can contain a smartcard operating system, and further in its memory contain an applet which is configured to generate a digital tag. The smartcard can also contain a communication medium, such as an NFC medium. The application launched on a user device (e.g. user device **200**) can then, based on the user input cause the user device to send commands to the smartcard to disable certain digital tags to be generated. In another example, the smartcard can contain a flag which can disable certain digital tags from being generated. Step **630** can consist of further configuring the application, once launched in a visible state, to modify the applet generating the digital tags on the first device. This modification can be made to modify the particular payloads of the digital tags which are generated on the first device.

(53) FIG. **7** shows a block diagram of an exemplary embodiment of a system **700** according to the

present disclosure, which can be used to perform the procedures described below. For example, exemplary procedures in accordance with the present disclosure described herein can be performed by a processing arrangement and/or a computing arrangement (e.g., computer hardware arrangement) **705**. Such processing/computing arrangement **705** can be, for example entirely or a part of, or include, but not limited to, a computer/processor **710** that can include, for example one or more microprocessors, and use instructions stored on a computer-accessible medium (e.g., RAM, ROM, hard drive, or other storage device).

(54) As shown in FIG. 7, for example a computer-accessible medium **715** (e.g., a storage device such as a hard disk, floppy disk, memory stick, CD-ROM, RAM, ROM, etc., or a collection thereof) can be provided (e.g., in communication with the processing arrangement **705**). The computer-accessible medium **715** can contain executable instructions **720** thereon. In addition, or alternatively, a storage arrangement **725** can be provided separately from the computer-accessible medium **715**, which can provide the instructions to the processing arrangement **705** so as to configure the processing arrangement to execute certain exemplary procedures, processes, and methods, as described herein above.

(55) Further, the exemplary processing arrangement **705** can be provided with or include one or more input/output ports **735**, which can include, for example a wired network, a wireless network, the internet, an intranet, a data collection probe, a sensor, etc. or any combination thereof. As shown in FIG. 7, the exemplary processing arrangement **705** can be in communication with an exemplary display arrangement **730**, which, according to certain exemplary embodiments of the present disclosure, can be a touch-screen configured for inputting information to the processing arrangement in addition to outputting information from the processing arrangement, for example. Further, the exemplary display arrangement **730** and/or a storage arrangement **725** can be used to display and/or store data in a user-accessible format and/or user-readable format.

(56) Throughout the specification and the claims, the following terms take at least the meanings explicitly associated herein, unless the context clearly dictates otherwise. The term “or” is intended to mean an inclusive “or.” Further, the terms “a,” “an,” and “the” are intended to mean one or more unless specified otherwise or clear from the context to be directed to a singular form.

(57) In this description, numerous specific details have been set forth. It is to be understood, however, that implementations of the disclosed technology may be practiced without these specific details. In other instances, well-known methods, structures and techniques have not been shown in detail in order not to obscure an understanding of this description. References to “some examples,” “other examples,” “one example,” “an example,” “various examples,” “one embodiment,” “an embodiment,” “some embodiments,” “example embodiment,” “various embodiments,” “one implementation,” “an implementation,” “example implementation,” “various implementations,” “some implementations,” etc., indicate that the implementation(s) of the disclosed technology so described may include a particular feature, structure, or characteristic, but not every implementation necessarily includes the particular feature, structure, or characteristic. Further, repeated use of the phrases “in one example,” “in one embodiment,” or “in one implementation” does not necessarily refer to the same example, embodiment, or implementation, although it may.

(58) As used herein, unless otherwise specified the use of the ordinal adjectives “first,” “second,” “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to, and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking, or in any other manner.

(59) While certain implementations of the disclosed technology have been described in connection with what is presently considered to be the most practical and various implementations, it is to be understood that the disclosed technology is not to be limited to the disclosed implementations, but on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

(60) This written description uses examples to disclose certain implementations of the disclosed technology, including the best mode, and also to enable any person skilled in the art to practice certain implementations of the disclosed technology, including making and using any devices or systems and performing any incorporated methods. The patentable scope of certain implementations of the disclosed technology is defined in the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A device, comprising: a processor; a memory storing an application; and a communication interface configured to generate a communication field, wherein, after receipt of a digital tag from a transmitting device after entry of the transmitting device into the communication field, the application is configured to: record a number of application launches responsive to the digital tag, and send, via the communication interface and a network, a notification to a server indicating the number of launches to synchronize a counter on the transmitting device with a counter on the server.
2. The device of claim 1, wherein: the communication interface is configured to generate the communication field using at least one selected from the group of near field communication (NFC), Bluetooth, Wi-Fi, a wireless local area network (WLAN), and radio-frequency identification (RFID), and the communication interface is in data communication with the server via at least one selected from the group of a wired network and a wireless network.
3. The device of claim 1, wherein the application is further configured to launch in a special state when the second device is in a predetermined state.
4. The device of claim 3, wherein the special state comprises a non-visible state.
5. The device of claim 3, wherein the predetermined state comprises at least one selected from the group of a locked state, an unlocked state, a screen on state, and a screen off state.
6. The device of claim 3, wherein: the device further comprises a battery, and the predetermined state is based on the battery.
7. The device of claim 3, wherein the predetermined state is based on power available to the device.
8. The device of claim 3, wherein the predetermined state is based on a time threshold between a first action and a second action.
9. The device of claim 8, wherein: the device further comprises a screen, the first action comprises an unlock of the screen, and the second action comprises receipt of the digital tag.
10. The device of claim 1, wherein the application is configured to launch when the device is at least one of a list of predetermined states.
11. The device of claim 1, wherein the list of predetermined states comprises at least one selected from the group of a locked state, an unlocked state, a screen on state, and a screen off state.
12. A method, comprising: receiving, by an application stored in a memory of a device comprising a processor and a communication interface from a transmitting device after entry of the transmitting into a communication field generated by the communication interface, a digital tag; recording, by the application, a number of launches responsive to the digital tag; and sending, by the application via the communication interface and a network, a notification to a server indicating the number of application launches to synchronize a counter on the transmitting device with a counter on the server.
13. The method of claim 12, further comprising intercepting, by the application, additional information from the transmitting device when the device is in at least one state from a list of predetermined states.

14. The method of claim 13, wherein the list of predetermined states comprises at least one selected from the group of a locked state, an unlocked state, a screen on state, and a screen off state.
 15. The method of claim 13, wherein the additional information from the transmitting device is intercepted for a fixed time duration after receipt of the digital tag.
 16. The method of claim 13, further comprising storing, by the application, the additional information in the memory.
 17. A non-transitory computer-accessible medium comprising instructions for execution by a computer hardware arrangement, wherein, upon execution, the instructions cause the computer hardware arrangement to perform procedures comprising: receiving, from a transmitting device after entry of the transmitting device into a communication field generated by a communication interface of the computer hardware arrangement, a digital tag; recording a number of launches responsive to the digital tag; and sending, via the communication interface and a network, a notification to a server indicating the number of application launches to synchronize a counter on the transmitting device with a counter on the server.
 18. The non-transitory computer-accessible medium of claim 17, the procedures further comprising: transmitting a modification to an applet of the transmitting device configured to generate the digital tag, and the modification comprises at least one selected from the group of a change to a digital tag payload and disabling the generation of digital tags.
 19. The non-transitory computer-accessible medium of claim 17, the procedures further comprising intercepting additional information from the transmitting device when the computer hardware arrangement is in at least one state from a list of predetermined states.
 20. The non-transitory computer-accessible medium of claim 19, wherein the list of predetermined states comprises at least one selected from the group of a locked state, an unlocked state, a screen on state, and a screen off state.
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